

7. The table shows some properties of elements in Period 3 of the Periodic Table.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Appearance	Malleability	Conductivity
Na	98	882	1.00	shiny solid	malleable	good
Mg	650	1091	1.75	shiny solid	malleable	good
Al	660	2470	2.70	shiny solid	malleable	good
Si	1410	3265	2.35	shiny solid	brittle	semiconductor
P	44	281	1.80	red solid	brittle	poor
S	113	444	2.05	yellow solid	brittle	poor
Cl	−101	−34	0.003	green gas	n/a	poor

(a) One of the elements is difficult to classify as a metal or non-metal.

Identify this element and give your reasoning.

[2]

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- (b) Tick (✓) **two** boxes which correctly describe the change in density and boiling point for the elements across Period 3. [2]

The density of metals and non-metals increases

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The boiling point of metals increases but the boiling point of non-metals shows no trend

☐

The density of metals shows no trend but the density of non-metals decreases

☐

The boiling point of metals and non-metals shows no trend

☐

The density of metals increases but the density of non-metals shows no trend

☐

The boiling point of metals shows no trend but the boiling point of non-metals decreases

☐

The density of metals decreases but the density of non-metals shows no trend

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- (c) Argon is the next element in Period 3 after chlorine, Cl.

State why it is not possible to predict a melting point for argon using the information in the table. [1]

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(d) Phosphorus is found in phosphoric acid, H_3PO_4 .

(i) During the production of phosphoric acid, phosphorus is heated to 60°C .

Give the state of phosphorus at 60°C . Explain your choice.

[2]

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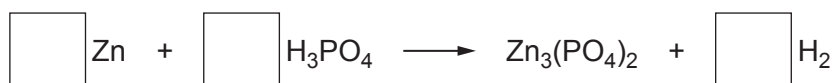
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(ii) Phosphoric acid reacts with zinc to produce zinc phosphate and hydrogen.

Balance the equation for this reaction.

[1]



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